

GATEMATE 20MM PVC SYSTEM

Official Setup Guide, Unique Connector Names & Launch Blueprint (v5)

This comprehensive manual acts as the single source of truth for the **GateMate 20mm System**. Engineered for FPV drone racing tracks or Fresstyle obstacles , every connector features a distinct, unique product code and name. All components are optimized with a flat bottom profile for 100% support-free FDM 3D printing and seamless bed adhesion.

1. The Named Connector Ecosystem

The GateMate catalog consists of six distinct, print-optimized hardware components. Each features a built-in 1.3% volumetric compensation factor to snugly grip standard 20mm OD PVC conduits via a durable friction fit without requiring messy adhesives.

Product Code & Name	Functional Design Description	Bed Alignment
GM-L2D "Corner Flat"	Standard horizontal 2-way 90-degree corner block. Restrained to a single plane.	100% Flat Base
GM-T2D "Split Flat"	Planar 3-way perpendicular T-junction. Splits pathways along the printing plane.	100% Flat Base
GM-X2D "Cross Flat"	4-way intersection mapping a complete flat coordinate layout on a single plane.	100% Flat Base
GM-L3D "Anchor Cube"	A 90-degree corner piece that incorporates an additional vertical column channel lifting directly off the center.	100% Flat Base
GM-T3D "Spire Base"	A planar T-junction featuring a standalone vertical riser sleeve directly centered above the core block.	100% Flat Base
GM-X3D "Matrix Hub"	A full 4-way horizontal cross grid with an integrated vertical riser stack pointing straight along the Z-axis.	100% Flat Base

Printability Guideline: All prints are design to be printed without supports. Make sure your bed is leveled because even the slighes amount of warping makes the fit very bad. A layerhight of 0.28mm and 3-5 wall loops are recommended for 5 inch 2 wall loops are more than enough for tinywhoops.

2. Tooling & Pipe Preparation Protocol

Clean pipe treatment is necessary to protect the internal plastic lining of the connectors from degradation over repeated track build cycles.

- **Recommended Cutting Tool:** A ratchet-action plastic pipe shears tool to achieve precise 90-degree cuts. Avoid raw wood saws as they fray the plastic.
- **Mandatory Beveling:** Every pipe cut must be actively sanded down with medium grit sandpaper. Polish the outer sleeve lip at a 45-degree angle to give it a smooth chamfered nose.

CRITICAL SYSTEM ALERT: Raw, un-sanded PVC pipe cuts act as razor-sharp cutting tools. Pushing un-beveled edges into the joints will slice away the inner printed friction-fit teeth of your GateMate connectors, permanently stripping out their holding tension.

3. Standard Field Layout Configurations (more coming soon...)

Using our designated product lineup, the field crew can assemble and breakdown tracks rapidly without field tools.

Config A: The "Ceiling-Hung Square Gate" (Suspended Setup)

This configuration utilizes no floor parts and cannot stand on its own. It is designed to be suspended from overhead rafters, branches, or structural ceiling grids.

Hardware Inventory	PVC Conduit Cutting Blueprint	Final Dimensions
4x GM-L2D "Corner Flat"	4x 80 cm conduit lengths	Outer: ~84 x 84 cm Inner: ~76 x 76 cm

Config B: The "Self-Standing Ground Square Gate" (Floor Setup)

The standard floor-based element for open surfaces. It anchors directly to gym floors or fields by employing outrigger stabilization extensions to prevent blowouts from micro-drone air currents.

Hardware Inventory	PVC Conduit Cutting Blueprint	Final Dimensions
2x GM-L2D "Corner Flat" 2x GM-T2D "Split Flat"	Main Frame: 1x 80 cm (Top Header Beam) 2x 80 cm (Vertical Side Studs) Stabilizing Foot Outriggers: 4x 15 cm (Floor Base Shoes; 2 per side)	Outer Frame: ~84 x 84 cm Inner Clearance: ~76 x 76 cm Foot Depth: ~34 cm

Config B Assembly Instruction: Position the two GM-T2D connectors flat on the floor to serve as the structural feet. Slide your vertical 80 cm frame side studs directly into the top socket of each GM-T2D. Finally, insert two 15 cm pipe elements horizontally into each side to create a balanced, tip-proof outrigger base facing front and back.

4. Pricing & Licensing Architecture

Commercial Tier	Scope of Assets Supplied	MSRP / Terms
Digital - Personal Use	Full 6-piece print-optimized STL package (GM-L2D to GM-X3D) for private single-user track setups only.	€4.99
Digital - Commercial License	Full 6-piece named STL catalog with commercial distribution rights or manufacturing authorization for racing leagues and commercial organizers.	Inquire via Email
Physical - Starter Kit	Pre-ordered, fully printed 10-piece package (6x GM-L2D, 4x GM-T2D) printed in robust 5-wall PLA. (Pipes excluded).	Inquire via Email

Commercial and Physical Inquiries: Commercial licenses and pre-printed physical kits are not sold automatically. For custom pricing structures, volume requests, or commercial usage permissions, you must contact the creator directly at: maximisslik1@hotmail.com.